

Unified Modeling Language(UML) Overview

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Software Modeling

model

- **Model = Abstraction of the system** 시스템의 추상화
 - Modeling is a way to deal with ~~complexity~~ by ~~ignoring irrelevant details~~ ⇒ 복잡한 걸 비필요적인 디테일을 제거함으로써 다룸.
 - Share idea and the results of system analysis and design among clients, system analysts, and programmers
- **Unified Modeling Language (UML)**
 - de facto standard for modeling ↳ SW 시스템을 시각적으로

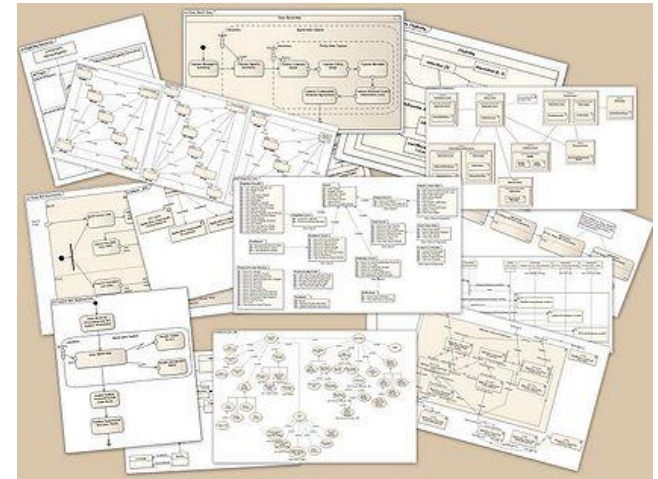
↳ SW system을 시각적으로 모델링하기 위한 표준화 표기법. 모델링하기 위한 표준화 표기법



System



Abstraction



Model in UML



UML History

“Method Wars” of object-oriented paradigm in 1990s

- OOAD, OMT, OOSE, Fusion, OOA/OOD
- Each of the methods had its own notation, process, and tools

UML = OMT by Rumbaugh
+ OOAD by Booch
+ OOSE by Jacobson
+ ...

Syntax(구문): 기호의 모양, 형태, 표기방식
Semantics(의미): 기호가 무엇을 뜻하는지 해석.

문법은 같은데, 의미가 다른 방법함.

Currently UML 2.5.1 (Dec 2017,
<http://www.omg.org/spec/UML/About-UML>)

2005: UML 2.0

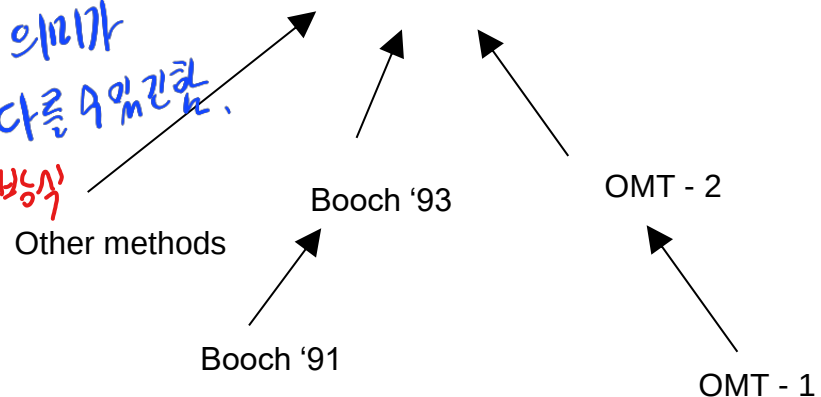
2001: UML 1.4

1999: UML 1.3

1997: UML 1.0, 1.1

1996: UML 0.9 & 0.91

1995: Unified Method 0.8



Categories of UML diagrams

- **Structure Diagrams** : what things must be in the system

- Class diagram
- Component diagram
- Composite diagram
- Deployment diagram
- Object diagram
- Package diagram

↳ 반드시 존재해야 하는 거

- **Behavior Diagrams** : what must happen in the system

- Activity diagram
- State diagram
- Use case diagram

↳ 사건 발생

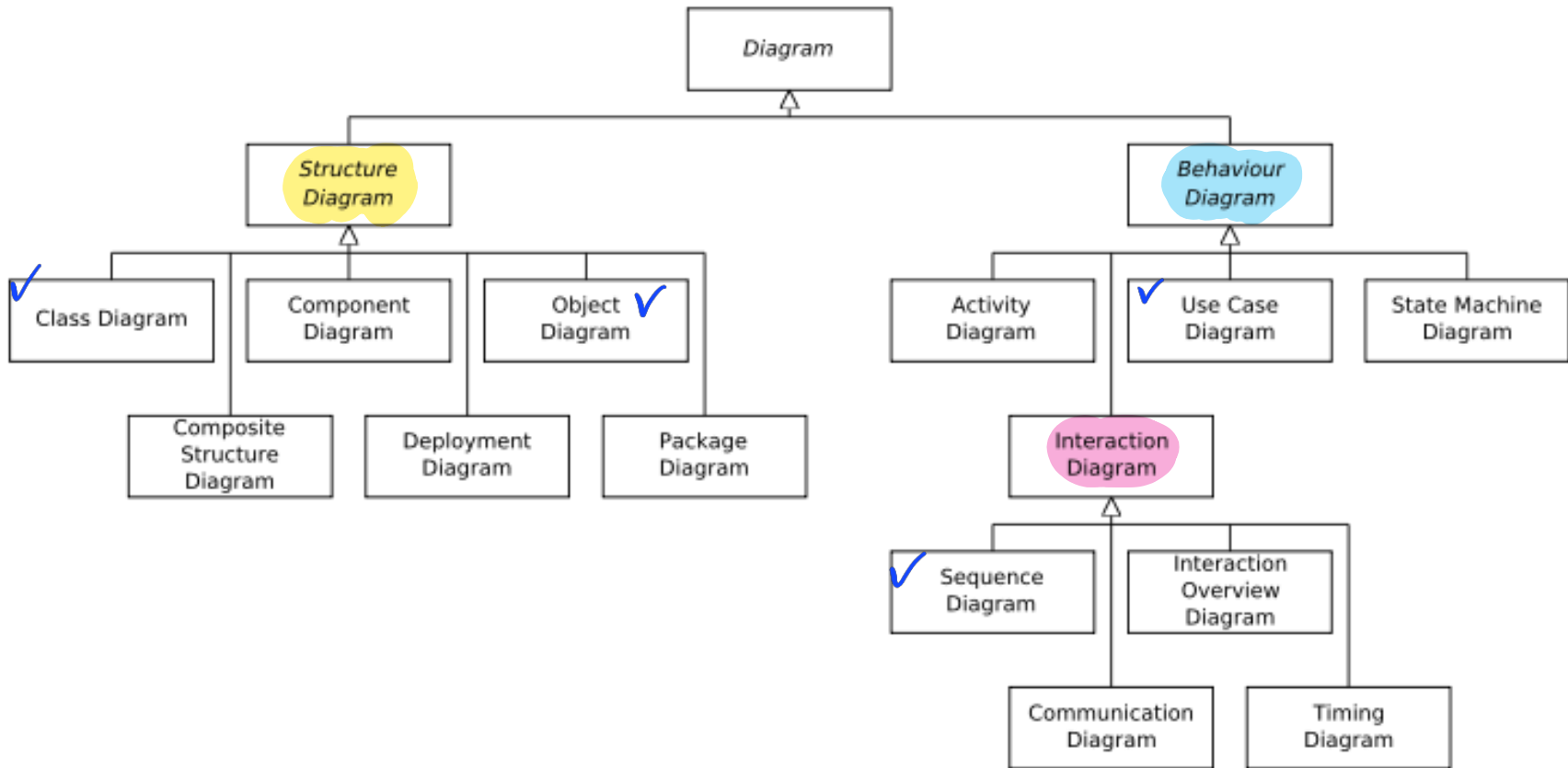
- **Interaction Diagram** (subset of behavior diagram) : the flow of control and data among the things

- Communication diagram
- Interaction overview diagram
- Sequence diagram
- Timing diagram

객체와
상호작용과 관련
간의 제어 + 데이터 흐름.



Categories of UML diagrams

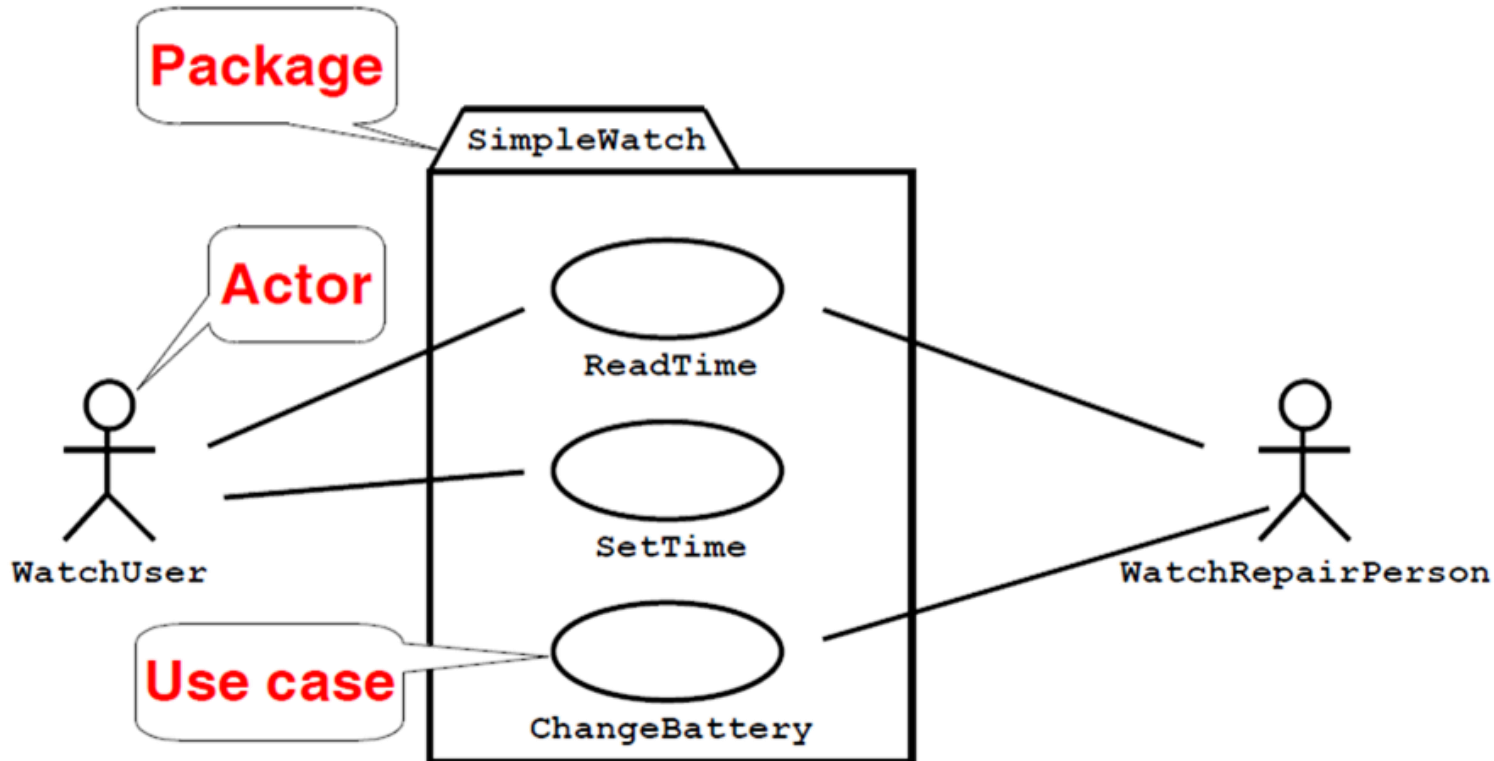


Use-Case Diagram

- shows the functions provided by a system in terms of

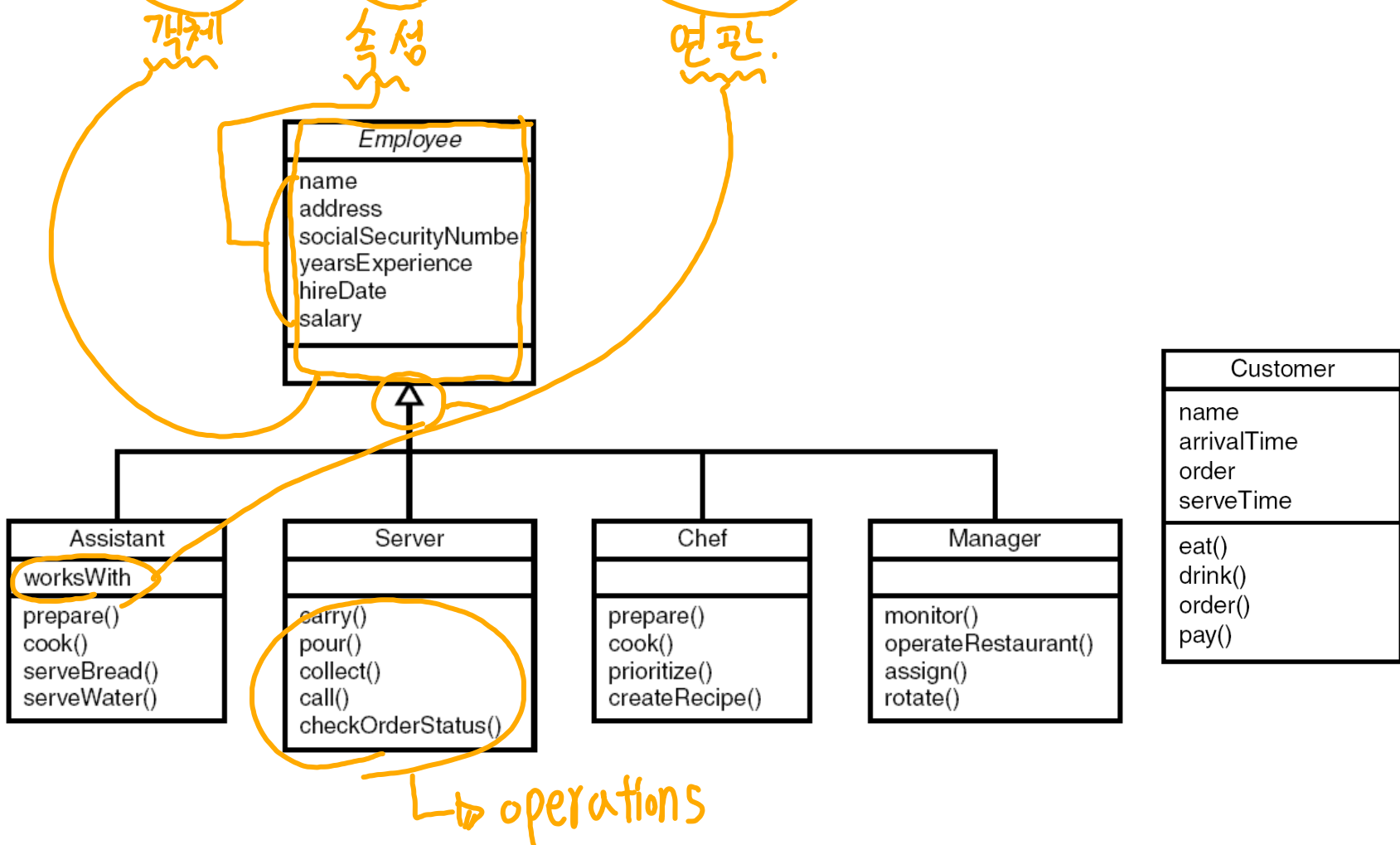
- Actors,
- Their goals represented as use cases, and
- Dependencies between those use cases

⇒ 기능들을 Actor, 목표, 의존성 관점에서 보여줌.



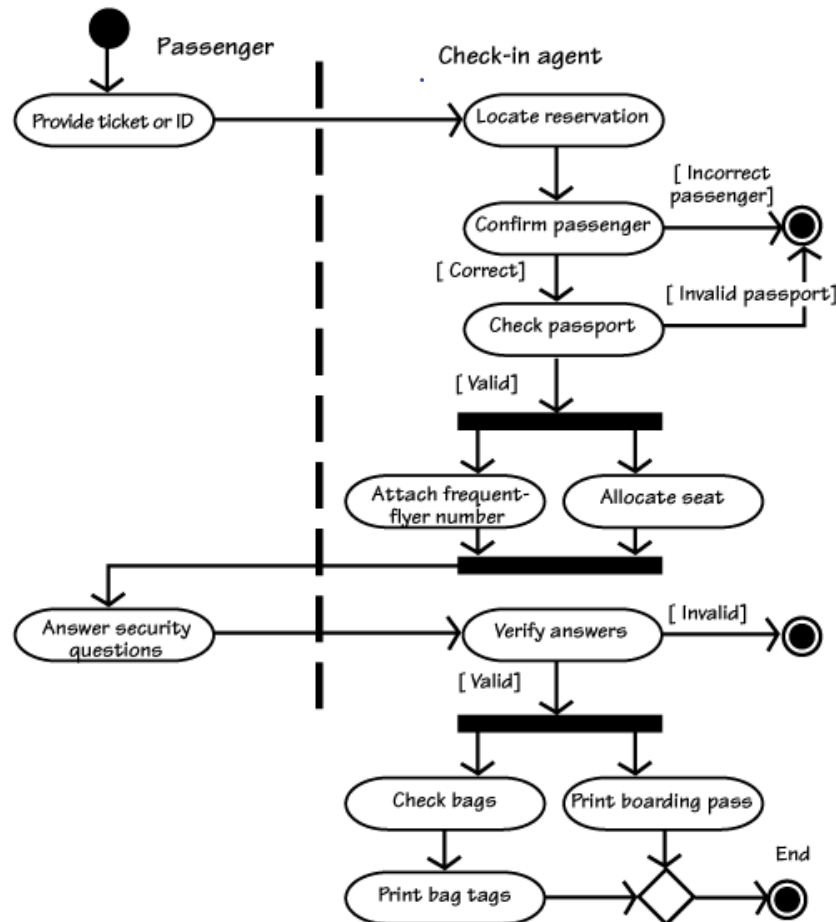
Class Diagram and Object Diagram

- Describe the **static structure** of the system ⇒ 정적 구조 묘사
- Objects**, **attributes**, and **associations**



Activity Diagram

- Step-by-step workflows
- Shows **overall flow of control** (flowchart)

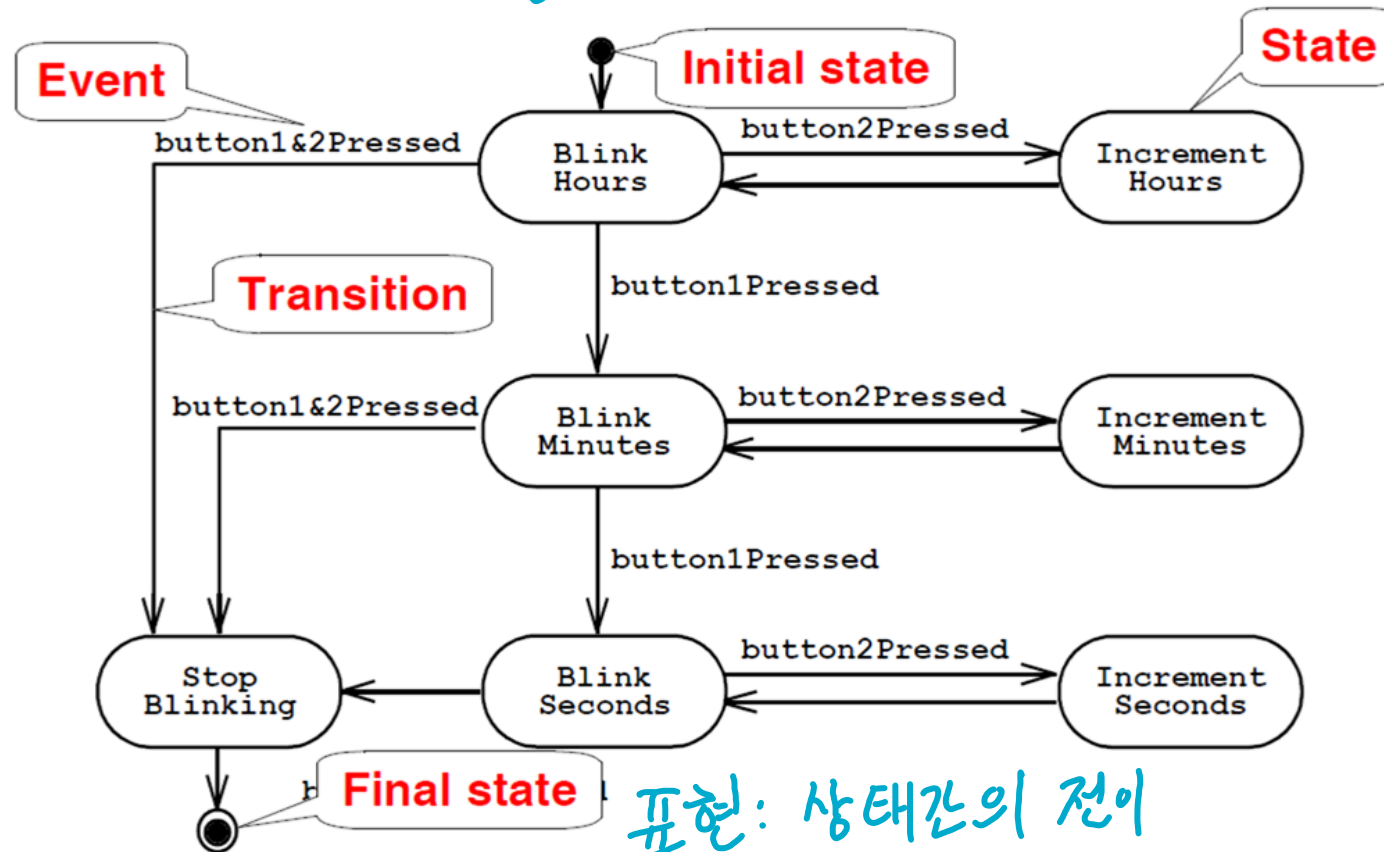


State (Statechart) Diagram

- Describe the **dynamic behavior** of an **individual object** as a finite state machine.

↳ 상태 변화를 보여줌.

↳ 개별 객체

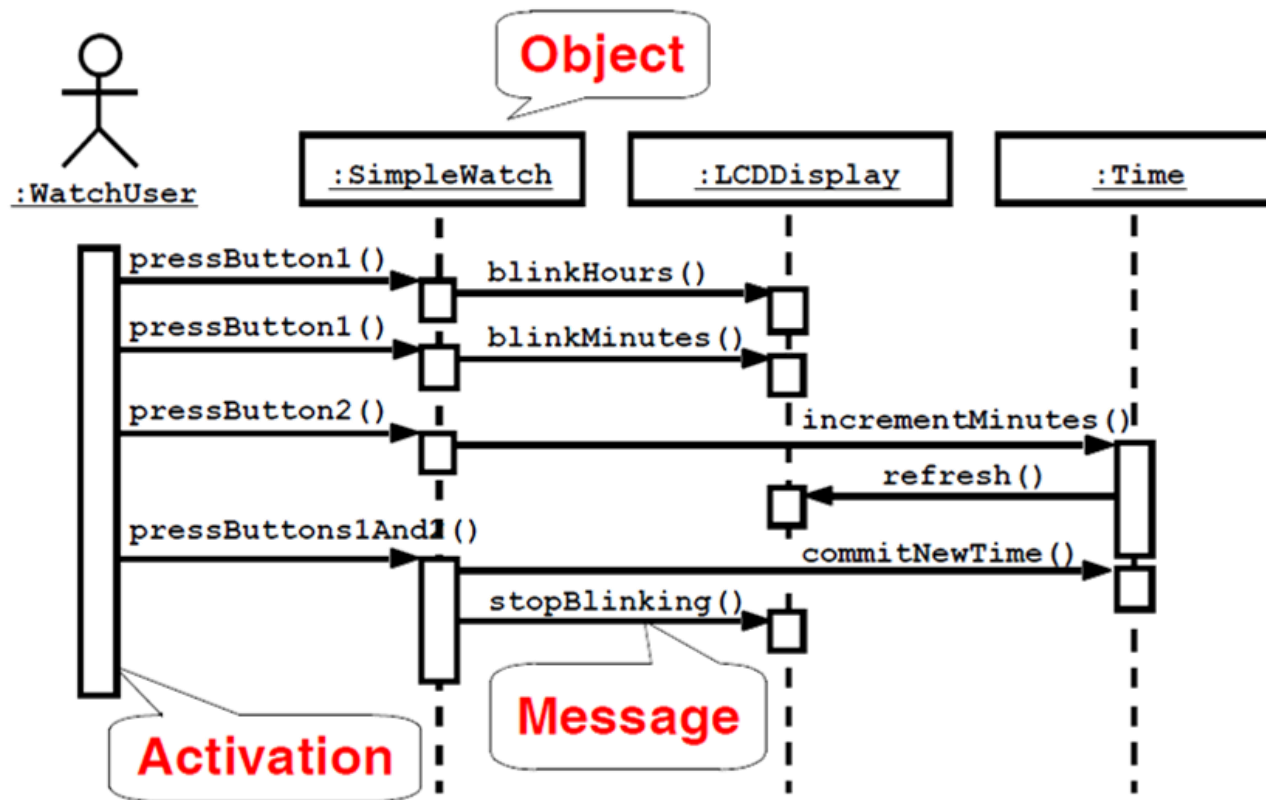


표현: 상태간의 전이
트리거: 이벤트 발생.

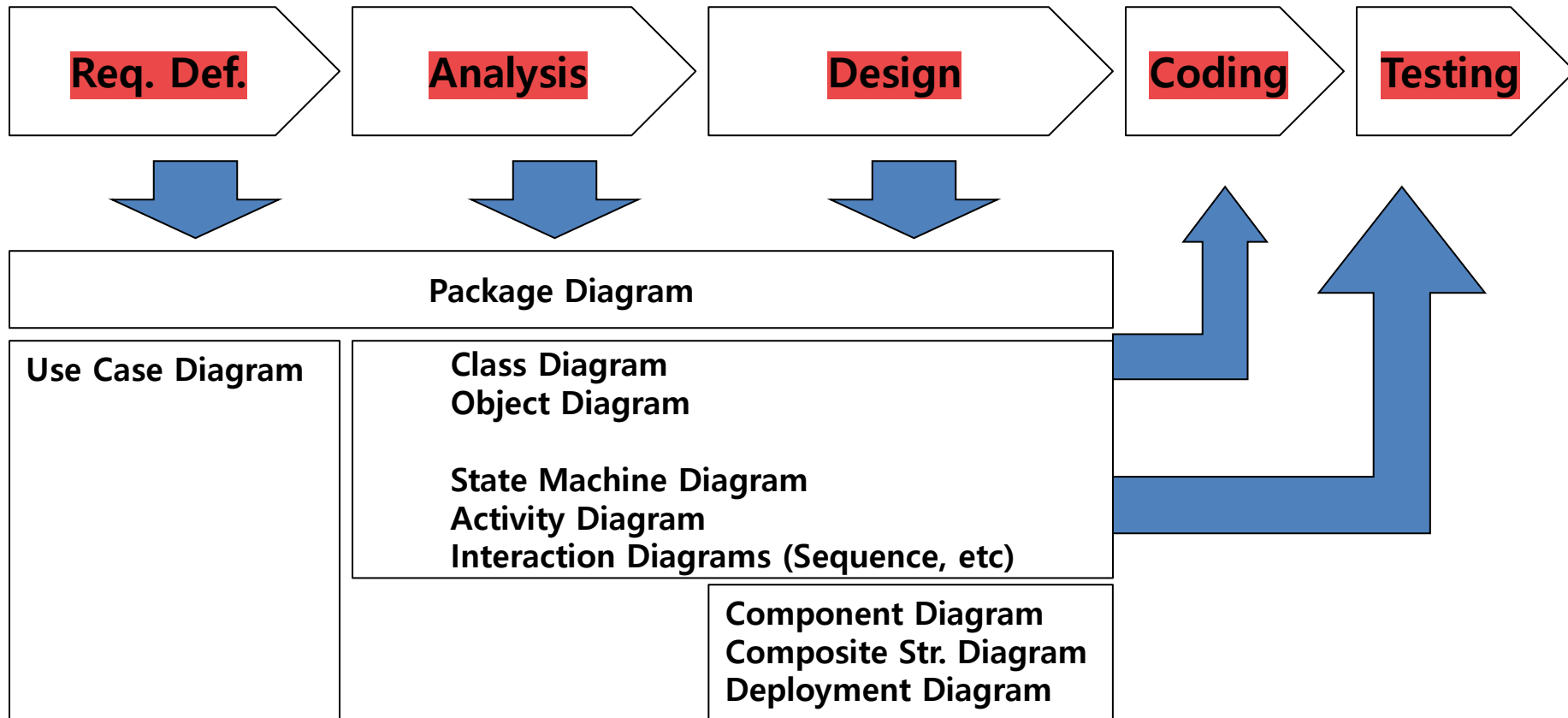


Sequence Diagram

- Describe the dynamic behavior
 - between actors and the system
 - between objects of the system.

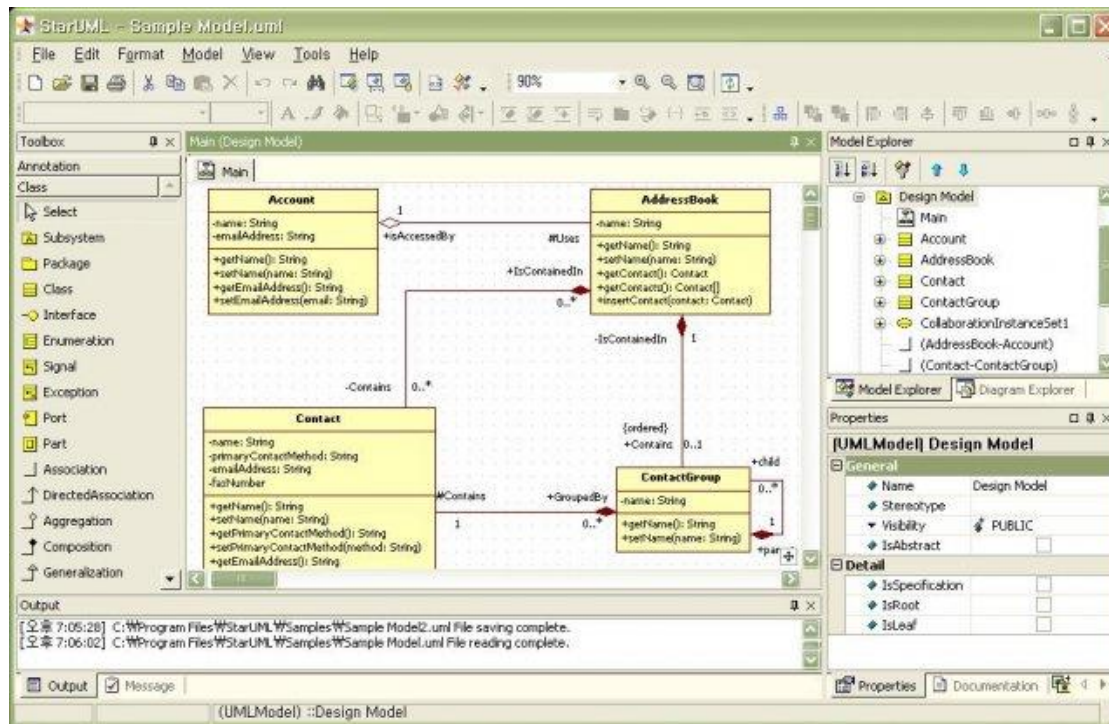


UML and Development Process



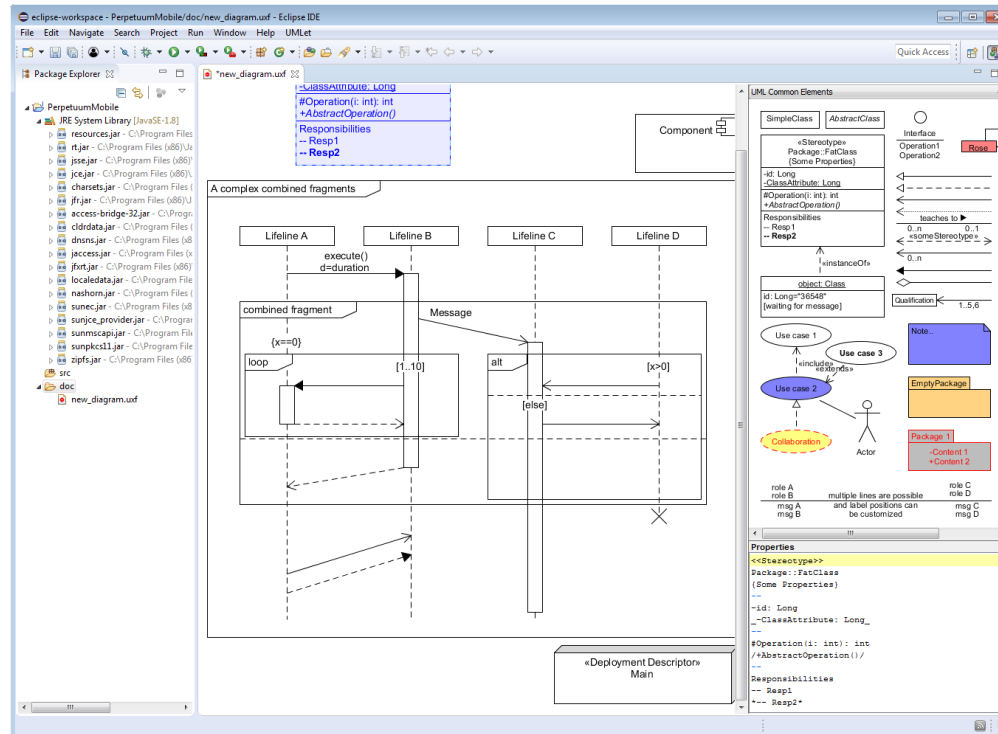
Free UML tool: StarUML

- Open source/platform
- Drawing diagrams, automatic code generation, ...
- <http://staruml.sourceforge.net/>



Free UML tool: UMLet

- Open source/platform
- Fast editing through text input
- Very good at class diagram
- <https://www.umlet.com/>



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Summary

- History of UML
- Structure Diagrams
- Behaviors Diagrams
- Interaction Diagrams
- Tools